

Nonnegative and positive factorizations

The nonnegative rank of the nine-point distance matrix

Difficulty: hard **Importance:** interesting to specialist

Status: open; checked 2026-09-08

Area: exact factorization of Euclidean distance matrices

Context and notation

All factorizations are over the real numbers. For $X \in \mathbb{R}_{\geq 0}^{m \times n}$, define

$$\text{rank}_+(X) = \min\{r \geq 0 : X = WH, \quad W \in \mathbb{R}_{\geq 0}^{m \times r}, H \in \mathbb{R}_{\geq 0}^{r \times n}\}.$$

Problem statement

Let $D \in \mathbb{R}_{\geq 0}^{9 \times 9}$ have entries $D_{ij} = (i - j)^2$ for $1 \leq i, j \leq 9$.

Question

Does there exist an exact factorization $D = WH$ with $W \in \mathbb{R}_{\geq 0}^{9 \times 6}$ and $H \in \mathbb{R}_{\geq 0}^{6 \times 9}$? Equivalently, is $\text{rank}_+(D) = 6$ or 7 ?

References

Baeckelant, Vandaele, and Gillis, *Computing Lower Bounds on the Nonnegative Rank via Non-Convex Optimization Solvers*, Appendix A.1, Table 6 and its final paragraph, expressly isolates this case. Pavel Hrubeš, *On the nonnegative rank of distance matrices*, Information Processing Letters **112** (2012), 457–461, supplies the upper-bound construction cited there.

Status evidence

The July 2026 revision records bounds $6 \leq \text{rank}_+(D) \leq 7$ and explicitly calls the exact value open. Targeted searches for the nine-point linear Euclidean distance matrix and subsequent nonnegative-rank results found no resolution. This is a named unresolved benchmark in the source, rather than an arbitrary specialization of a separately counted general problem.